Tokens and Context

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach budget fit together with source completeness.

Supply exact token rules and full cards, distinguish cases and preserve the optional attention strand.

Keep one habit: Ask what fact becomes unsupported when a passage is omitted and what an actual model run has established.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Account for fixed text, **evidence** and **reserved output** under a toy budget

For the compact core route, use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

By the end, I can budget context and preserve **evidence** needed to interpret a requested fact.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a next-**token** table is a valid probability distribution

Reference example, not the only acceptable wording: Teacher explanation: Entries are nonnegative and sum to 1. Not ready has mass 0.3+0.2 = 0.5, including busy and unknown.

No. Evaluate retrieval and claim support directly using sources and task-specific criteria.

▼ Materials and extension planning

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

Apply the connected idea: Attention as weighted information selection

For the compact core route, use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

Summation removes row order, so different ordered vectors may produce the same aggregate.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

A Distribution of Possibilities

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

Concept to explain

Models operate on **token**s within a limited **context window**.

Term	Meaning
token	A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.
tokenization	Converting text into the token units of a particular tokenizer.
context window	The bounded amount of token information a model can process together in a particular use.
truncation	Removing part of information, often because it exceeds a length or context limit.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach budget, evidence completeness and faithful use separately

> Keep main room,80-unit practice and pictured event records distinct.

State the main accounting and room evidence

For this lesson only, a fictional system allows 100 total **token** units for **input** plus **reserved output**. Its fixed instructions use 20 units, question 10 and **output** reserve 20. That leaves 50 units for **evidence**.

These are supplied counts, not word counts or a claim about any deployed model. Other systems can use different **input**/**output** accounting, so read their exact contract.

The question is: "Which room is approved for the robotics meeting, and what is its capacity?" **Evidence** A, a current approval notice, costs 30 units and names room B.

Evidence B, the current room register, costs 20 and gives room B capacity 12. **Evidence** C, an old approval notice, costs 25 and names room A.

Compare main evidence pairs

Evidence	Cost	What it supports
A: current approval	30	Approved room B
B: current register	20	Room B capacity 12
C: superseded approval	25	Old room A approval

A+B costs 50 and fits exactly. Together they support both requested claims. A+C costs 55 and exceeds the **evidence** budget, while C+B fits at 45 but supplies an outdated room decision.

A cheap set is not necessarily an adequate set. Preserve document IDs, versions and dates alongside content.

Preserve qualifiers and scope

A source may contain an exception after a general statement. Cutting off the last paragraph can reverse its practical meaning.

Selection should preserve the passages needed to interpret the requested fact, including scope, version and exceptions.

If complete relevant **evidence** cannot fit, retrieve a smaller authoritative passage or request clarification rather than silently remove a qualifying condition.

Including a document in context does not prove the generated answer uses it correctly. After generation, check each claim against its source.

A source’s instruction-like content also does not acquire authority just because it fits in the **evidence** budget.

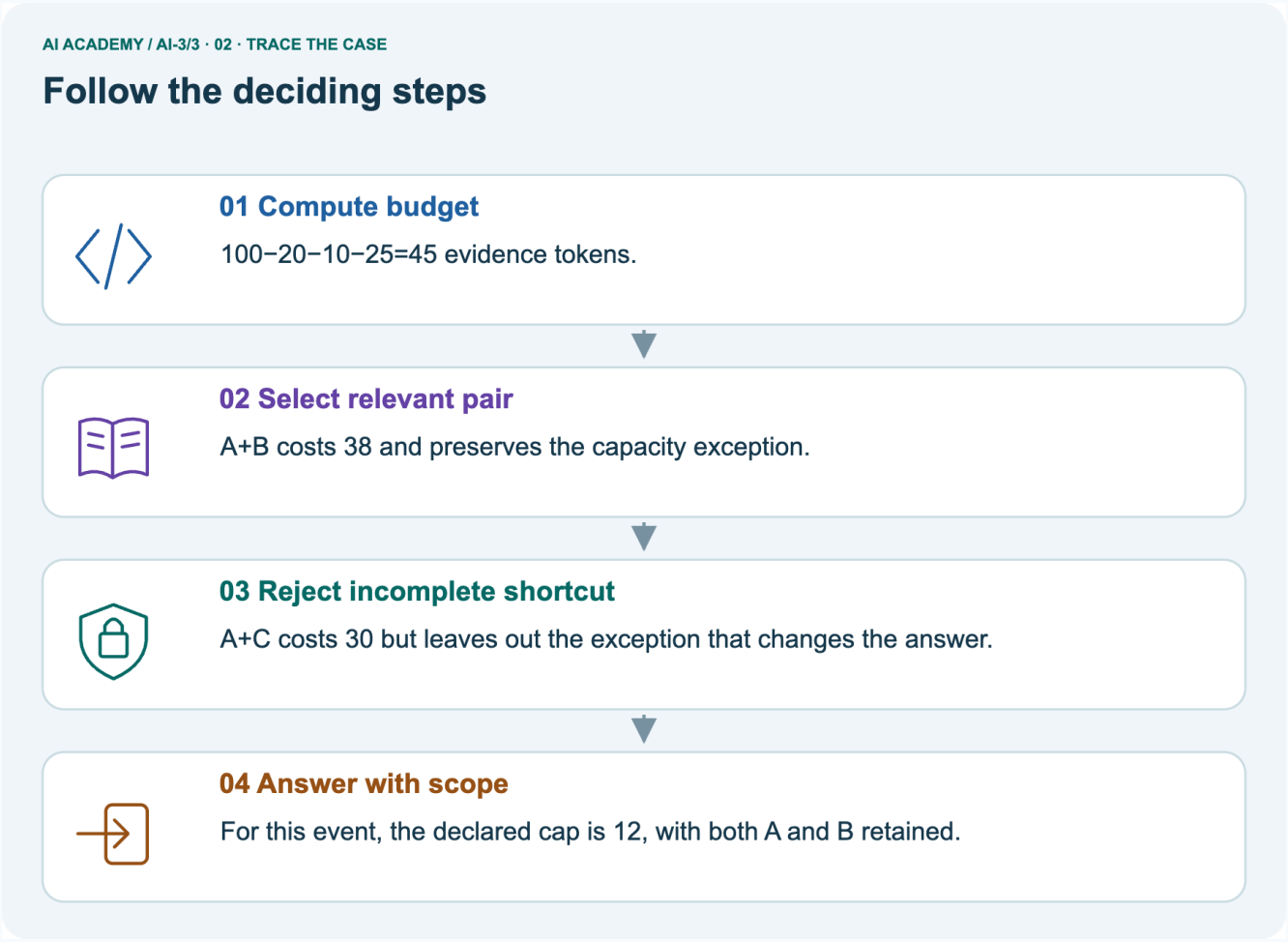
Read the separate event budget and exception

A fictional assistant has a small **context budget** for answering an event-capacity question.

Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Trace relevant support and exact costs



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute budget	100–20–10–25=45 evidence tokens.
2. Select relevant pair	A+B costs 38 and preserves the capacity exception.
3. Reject incomplete shortcut	A+C costs 30 but leaves out the exception that changes the answer.
4. Answer with scope	For this event, the declared cap is 12, with both A and B retained.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Predict the larger exception passage

Change: Increase B's cost to 30 with all other costs fixed.

Prediction: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Keep the evidence limit

Fitting in context does not guarantee attention, correct interpretation or accurate generation.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Attention extension · Mixture, masking and source-trust limit

Attention computes a weighted combination of **value** vectors, with weights derived from **query**-key compatibility. Queries, keys and **value**s are distinct roles even when they come from the same sequence. A weight is a numerical contribution in this computation, not proof of the source's truth or permission. This week supplies weights directly; Week4's retained attention extension calculates them from scores.

Use three **value**s V1=[2,0], V2=[0,4], V3=[4,2] with weights 0.5,0.25,0.25. **Value**s have equal width and weights are nonnegative with sum 1. Compute each coordinate separately. No **token** is literally copied by this combination.

Value	Weight	Weighted vector
[2,0]	0.5	[1,0]
[0,4]	0.25	[0,1]
[4,2]	0.25	[1,0.5]

Adding the weighted vectors gives [2,1.5]. The same weights applied to different **value**s can give a different result; the same **value**s with different weights can also change it. For example weights 0,1,0 yield V2=[0,4].

The result is a representation to be used by later operations, not automatically a generated word.

Guided: [6,0]/3 + 2*[0,3]/3 = [2,2]. **Mask**ing B leaves [6,0]. A **mask**ed source is excluded from the stated information path. Exit: coefficient size does not verify a document or grant instruction authority.

Primary reference for the optional attention strand: Vaswani et al., Attention Is All You Need, section3.2 defines attention using query/key compatibility and a weighted sum of values. Here the weights are supplied directly, and the empty-mask fallback is an explicit classroom contract. Reference: https://arxiv.org/html/1706.03762v7#S3.SS2

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. A sentence becomes **token** pieces, not a stored human thought.

2. The model uses visible **token**s to predict the next **token**.

3. Remove the definition from context: the same question becomes ambiguous.

4. More context helps only when it is relevant, accurate, and within the window.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A toy window holds eight **token**s; prompt uses six and quotation uses three.

Does it fit? What should be preserved?

Exact explanation to model

Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.
</details>

Explain the added worked example

Key terms

Term	Meaning
Context budget	The permitted input and output allocation under a stated model or classroom contract
Evidence selection	Choosing source material relevant to the requested claim
Reserved output	Capacity deliberately kept available for the response
Truncation	Removing part of a sequence to meet a limit

Apply the connected idea: Attention as weighted information selection

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Tokens and Context

The facts we will use

AI ACADEMY / AI-3/3

Context budget allocations

	Toy tokens
Instructions	20
Question	10
Output reserve	25
Evidence available	45

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the main explanation, then fade support

> Collect each learner’s reasoning before revealing a key.

Main completion question and key

Completion task: Compare A+B,A+C,C+B for fit and support of current approval and capacity.

Teacher check: Teacher explanation: A+B costs 50 and supports both. A+C costs 55 and fails budget; C+B costs 45 but uses old approval and fails current-approval support.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Main individual check

Use the worked case for Context limits and **evidence selection**. Proposed explanation: “**Token** count equals word count”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The lesson uses supplied counts; actual counts depend on **token**izer and protocol.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed case

Can A alone justify answering 20 seats after the exception is known?

No. The relevant exception limits this event to 12; omitting it would knowingly lose decisive **evidence**.

Changed-case reasoning: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Calculate a weighted combination of **value** vectors

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Calculate the **output** from the three supplied **value**s and weights.

Teacher check: Teacher explanation: Contributions [1,0],[0,1],[1,0.5] sum to [2,1.5].

Individual check: For Attention as weighted information selection, use its worked case to evaluate this proposed explanation: "Attention copies the highest-weight **token**". Explain the deciding step.

Expected correction: The supplied procedure combines vectors from multiple positions.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: A toy window holds eight **token**s; prompt uses six and quotation uses three.

Write the question to explain: Does it fit? What should be preserved?

Reveal after **prediction**s: Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.

Use the earlier transfer question as optional practice. Reserve Budget a new evidence set independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: 80-15-10-15 = 40; 18+22 fits. Claim support must still be checked. Exit: the register lists room B capacity 12, but current meeting approval is unavailable; request that approval before naming it as approved.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read the explanation in “Notice the evidence” and “See how it works”, then complete W1; ask for a reason alongside each answer.

Stage	Teacher action
18 to 32 minutes	Trace the “A closer explanation” material in “See how it works” and “Follow a worked example”, then complete W2. The pictorial case is an additional example.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read the explanation in “Notice the evidence” and “See how it works”, then complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the “A closer explanation” material in “See how it works” and “Follow a worked example”, then complete W2. The pictorial case is an additional example.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Attention as weighted information selection

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

Main evidence50: A+B50 supports both; A+C55 exceeds budget; B+C45 uses old approval. Pictured event evidence45: A+B38 preserves cap12; A+C30 omits it; Bcost30 makes A+B48 too large. Encoder: Ada!→[1,4], AI🙂 →[2,5], schoolbook→[8], school book→[6,3,7], ada fails a at0. Attention full mixture[2,1.5]; excludingV3 gives weights2/3,1/3 and output[4/3,4/3]; no permitted mass requires the declared fallback.

Independent D/E/F evidence50,D+E50fits,E+F40cannotreplacecurrentapproval.

Predict, investigate and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the supplied toy vocabulary/encoder and complete room-evidence cards. Predict exact **token** IDs, preserve unsupported-character failures, then compare size and claim support when a card is hidden. These are local calculations and paper traces, not measured model continuations.

4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

Token count equals word count: The lesson uses supplied counts; actual counts depend on **token**izer and protocol.

The smallest set is the best **evidence**: A set can fit while lacking current support for the requested claims.

Included means verified: Generation may misread or ignore **evidence**; verify claims after generation.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

Attention copies the highest-weight **token**: The supplied procedure combines vectors from multiple positions.

Deleting one contribution preserves a normalised mixture: Renormalise permitted mass under the stated contract.

Attention weights are truth probabilities: They are coefficients derived from compatibility, with no source-truth guarantee.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Budget a new evidence set independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

Teacher explanation: **Evidence** capacity 50. D+E costs 50 and supplies both current claims. E+F costs 40 but old approval is insufficient.

Without D, report the supported capacity and ask for current approval rather than infer it.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Compute available evidence

Under total 100, instructions 20, question 10 and **output** reserve 20, calculate **evidence** capacity.

Teacher explanation: 100-20-10-20 = 50 supplied **token** units.

Separate the total budget into its declared parts: 100 minus 20 instruction units minus 10 question units minus 20 **reserved output** units leaves 50 units for **evidence**. Preserve the **output** reserve before filling **evidence** space.

These are the lesson’s supplied **token** units, not a universal word-to-**token** conversion.

W2 Choose complete support

Compare A+B,A+C,C+B for fit and support of current approval and capacity.

Teacher explanation: A+B costs 50 and supports both. A+C costs 55 and fails budget; C+B costs 45 but uses old approval and fails current-approval support.

W3 Preserve a qualifier

A passage says room B normally holds 12, except during maintenance when the approved limit is 8. Why is retaining only the first clause inadequate?

Teacher explanation: The exception changes the applicable capacity during maintenance. Preserve conditions and determine whether maintenance applies; do not report 12 unconditionally.

W5 Audit inclusion

A team says every claim is verified because all chosen documents fit in context. What further check is needed?

Teacher explanation: Check each generated claim against the actual source passage/version and preserve qualifiers. Budget fit and inclusion do not establish faithful use or current correctness.

▼ Extension questions and keys

Attention as weighted information selection — W1 Identify the roles

Explain **query**, key, **value** and weight without calling the largest weight a truth score.

Teacher explanation: The **query** compares keys, compatibility produces weights, and weights combine **value**s. A weight is a computational coefficient, not a truth score.

Attention as weighted information selection — W2 Compute the mixture

Calculate the **output** from the three supplied **value**s and weights.

Teacher explanation: Contributions [1,0],[0,1],[1,0.5] sum to [2,1.5].

Multiply each coordinate by its **value**'s weight: 0.5*[2,0]=[1,0], 0.25*[0,4]=[0,1], and 0.25*[4,2]=[1,0.5]. Add first coordinates to get 1+0+1=2 and second coordinates to get 0+1+0.5=1.5.

The combined vector is [2,1.5], not a literal selection of one source **token**.

Attention as weighted information selection — W3 Mask and renormalise

Exclude V3 and calculate permitted weights and **output**. Explain the empty-**mask** case.

Teacher explanation: Retained total 0.75 gives weights 2/3,1/3,0 and **output** [4/3,4/3]. All-excluded gives zero total and must report no available **value** under this contract.

Attention as weighted information selection — W4 Combine a new case independently

Values A=[4,1],B=[0,5],C=[2,3]; weights 0.25,0.5,0.25. Compute the full **output**. **Mask** B, renormalise the remaining weights and compute the new **output**.

Teacher explanation: Full contributions [1,.25],[0,2.5],[.5,.75] give [1.5,3.5]. Retained total .5 gives A/C weights .5,.5 and **output** [3,2]. **Mask**ing changes available information; it does not verify sources.

Attention as weighted information selection — W5 Audit an explanation

A student claims the highest attention weight explains the entire model and authenticates a retrieved instruction. What additional reasoning is needed?

Teacher explanation: Inspect **value**s and downstream computations, other heads and residual paths before explaining the whole model.

Source truth and instruction authority require separate **evidence** and permissions; weight alone supplies neither.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.

Additional transfer practice

A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

Reasoned practice answer

The centimetres instruction may be outside the available context. Restate the required unit in the current request and check the **output**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “All earlier conversation remains available forever.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Total=14, exceeding the toy limit by 2. Select a relevant 5-**token** source extract or shorten other material while retaining essential instructions and **evidence**.

Real systems specify their own context and **output** limits; this toy policy must not be assumed universal.

Explanations for the additional workbook examples

W1 Compute available evidence

W2 Choose complete support

W3 Preserve a qualifier

W4 Budget new evidence independently

Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

Teacher explanation: **Evidence** capacity 50. D+E costs 50 and supplies both current claims. E+F costs 40 but old approval is insufficient.

Without D, report the supported capacity and ask for current approval rather than infer it.

W5 Audit inclusion

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

Main support and fresh assessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a **value** before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires 50, exact D+E fit, E+F inadequacy and an explicit missing-approval fallback. Fresh retry: total 90,instructions 20,question 10,reserve 20 gives 40; current approval costs 24 and capacity 16, exactly fit.

A cost-12 old approval does not replace current support.

Delayed return

Suggested return points in this level: ai-3/4, ai-3/6, ai-3/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 4: Sampling and Variability. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week4: Sampling and Variability. Carry forward the explicit vocabulary, conditional choices and **evidence** boundary when comparing repeated generated **output**s.

▼ Extension support and reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a **value** before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires [1.5,3.5] then [3,2] and explicit renormalisation. Fresh retry: A=[2,2], B=[6,0], C=[0,4], weights .5,.25,.25. Full **output** [2.5,2]; exclude A, remaining weights .5,.5 and **output** [3,2].

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Keep the learner’s question, approved source excerpt, and **output** rules inside the active context.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Attention as weighted information selection

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Account for fixed text, **evidence** and **reserved output** under a toy budget

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Compare A+B,A+C,C+B for fit and support of current approval and capacity.

Reasoned practice answer: Teacher explanation: A+B costs 50 and supports both. A+C costs 55 and fails budget; C+B costs 45 but uses old approval and fails current-approval support.

Check every learner before discussion: Use the worked case for Context limits and **evidence selection**. Proposed explanation: “**Token** count equals word count”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The lesson uses supplied counts; actual counts depend on **token**izer and protocol.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Increase B’s cost to 30 with all other costs fixed. Require a **prediction** and a reason before comparing. Expected result: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

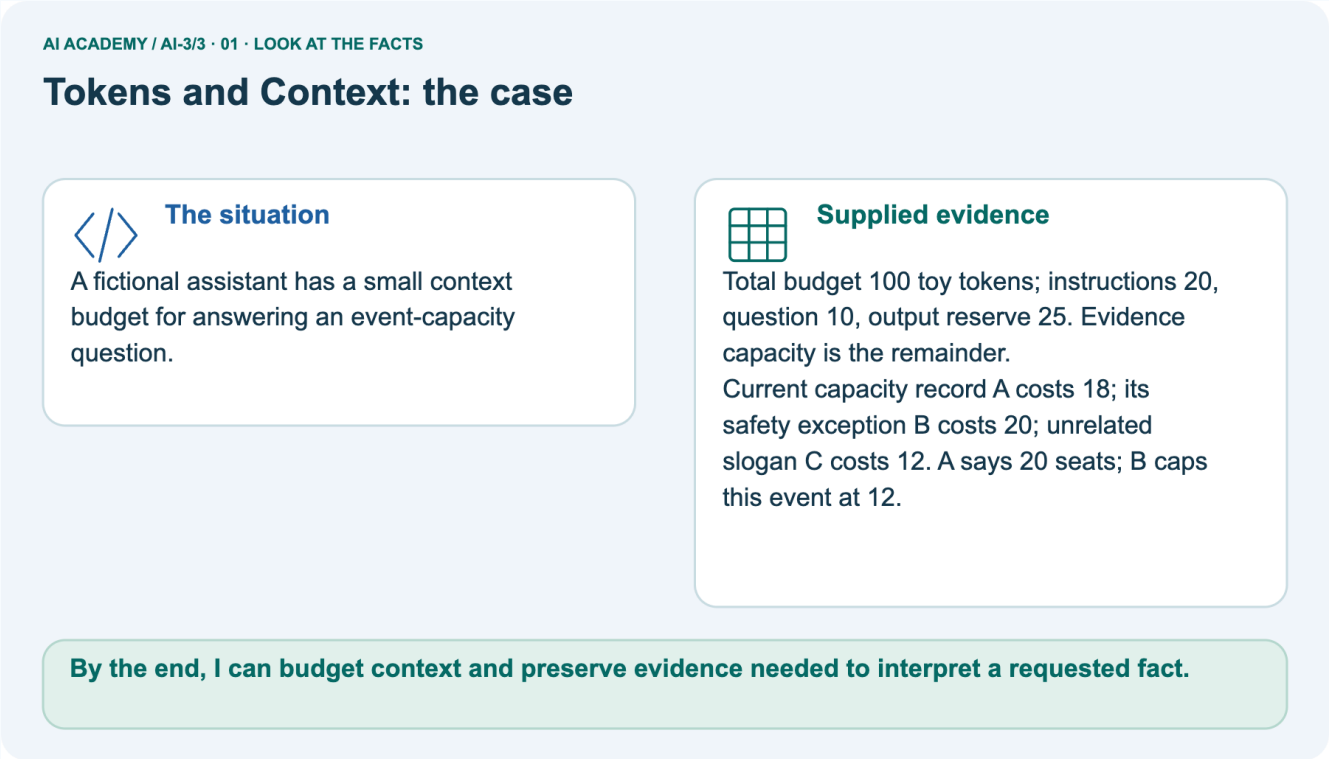
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Tokens and Context: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

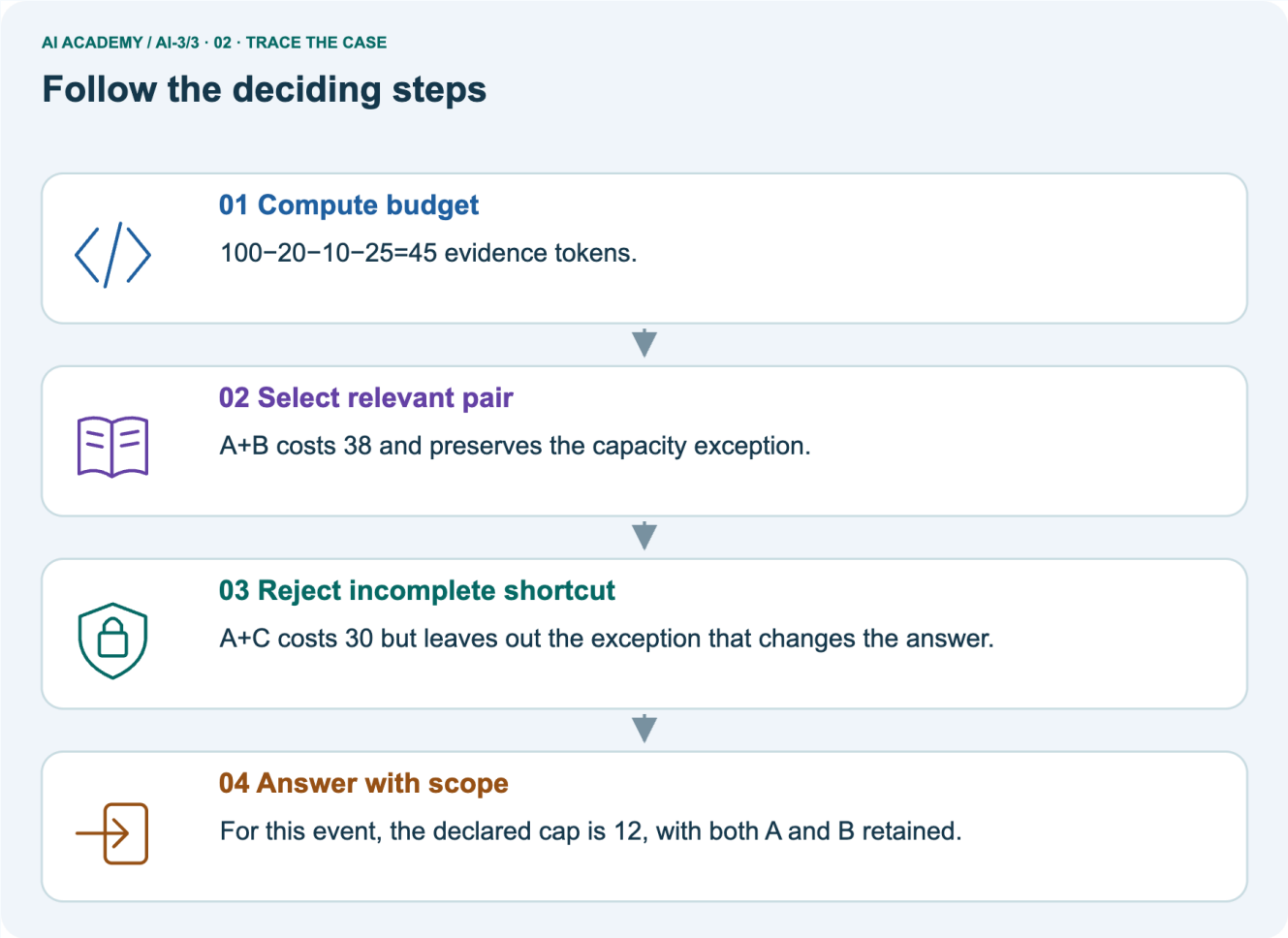
A fictional assistant has a small **context budget** for answering an event-capacity question. Supplied facts: Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute budget: 100−20−10−25=45 **evidence token**s. → Select relevant pair: A+B costs 38 and preserves the capacity exception. → Reject incomplete shortcut: A+C costs 30 but leaves out the exception that changes the answer. → Answer with scope: For this event, the declared cap is 12, with both A and B retained.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Context budget allocations



Follow the labelled picture with the matching explanation. Open the vector diagram

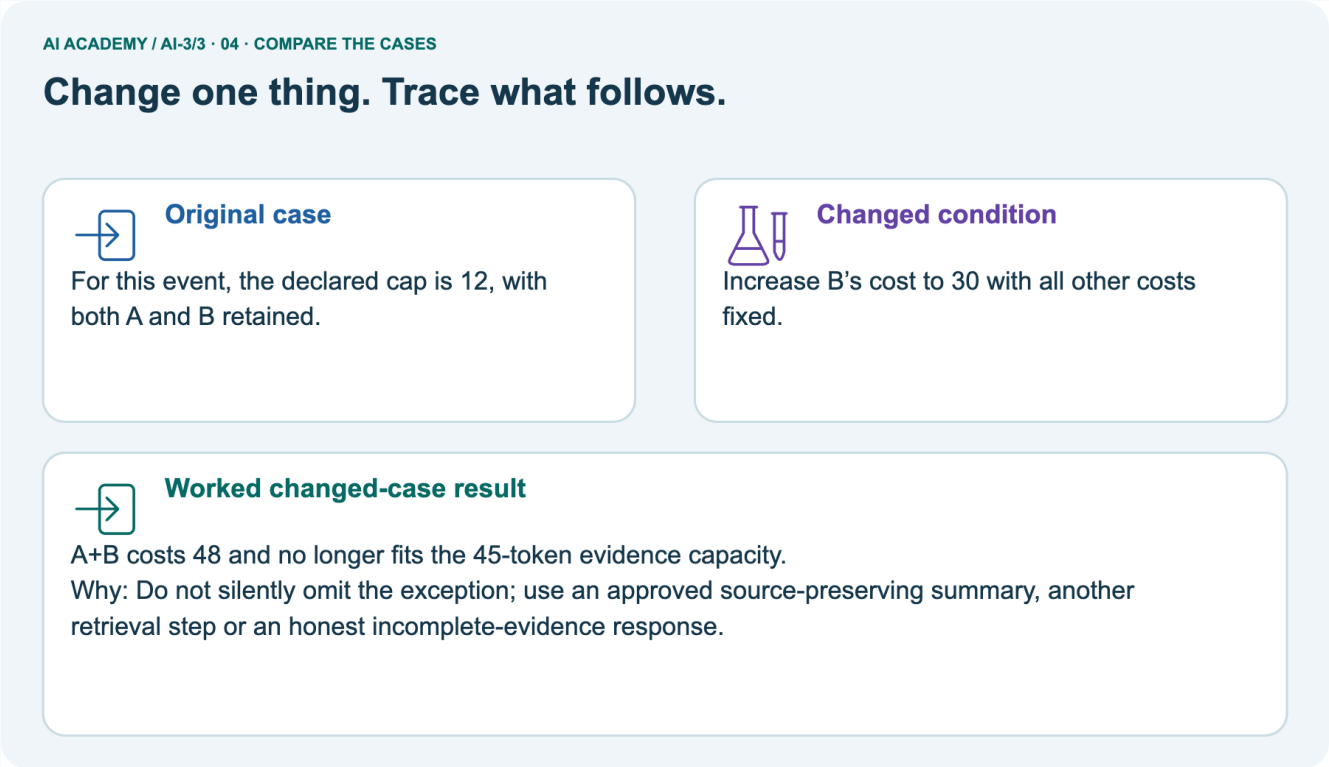
Context budget allocations. Instructions: Toy **token**s = 20. Question: Toy **token**s = 10. **Output** reserve: Toy **token**s = 25. **Evidence** available: Toy **token**s = 45.

Read the labels and **value**s. Explain the relationship the model shows.

Teacher prompt: Read the labels and **value**s. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy **value**s are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For this event, the declared cap is 12, with both A and B retained. Changed condition: Increase B's cost to 30 with all other costs fixed.

Expected result: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

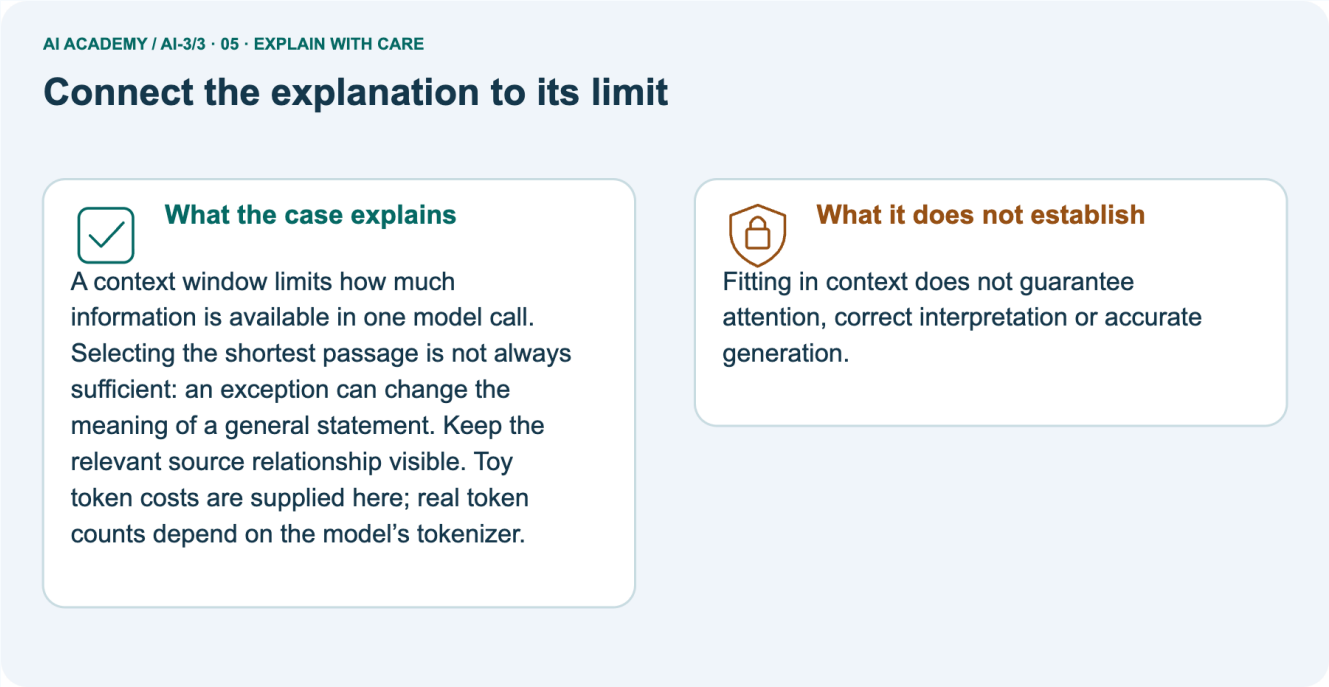
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see.

Changed-case reasoning: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

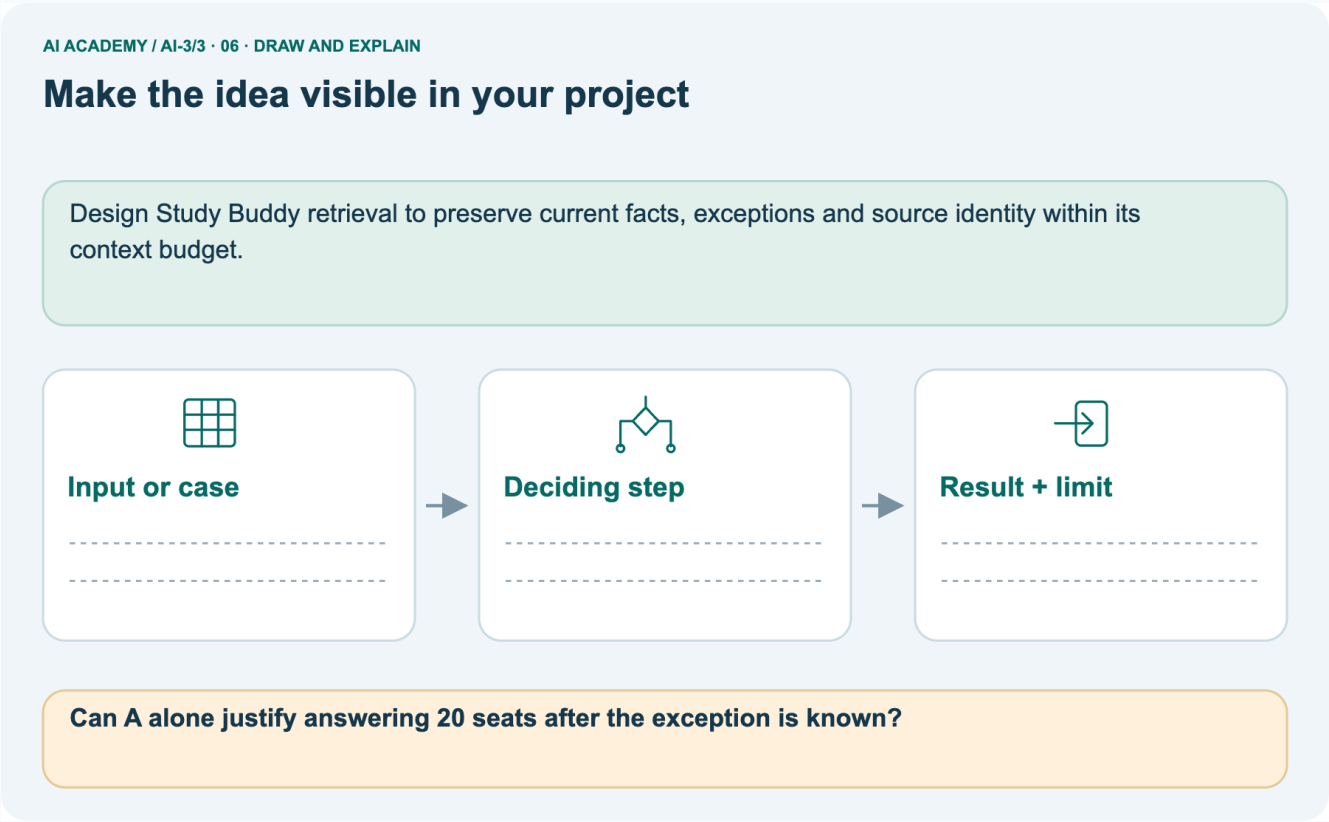
Limit: **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**. Draw your **input**, deciding step, result and limit.

Supported question: Can A alone justify answering 20 seats after the exception is known?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. The relevant exception limits this event to 12; omitting it would knowingly lose decisive **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Tokens and Context: the case

02. Follow the deciding steps

03. Context budget allocations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.